Separable Equations

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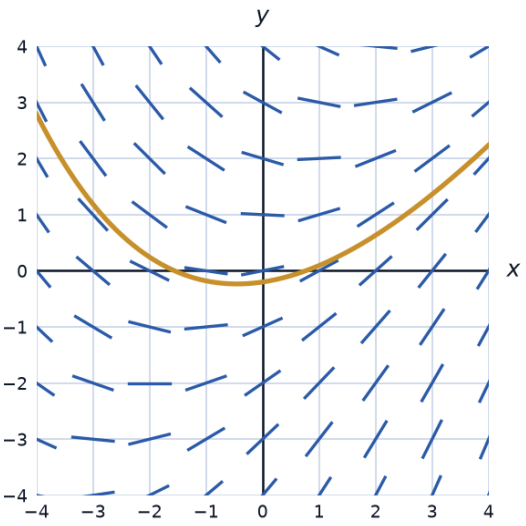
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Order and verifying solutions

State the order of the differential equation $y'' + 3y' - 2y = \sin x$.

2 [Foundational] • Target: Order and verifying solutions

State the order of $\left(\frac{dy}{dx}\right)^3 + xy = 0$, and note that order depends only on the highest derivative, not on any power of it.



3 [Foundational] • Target: Order and verifying solutions

Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.

4 [Foundational] • Target: Order and verifying solutions

Verify that $y = \frac{1}{x}$ is a solution of $y' + y^2 = 0$ for $x \neq 0$.

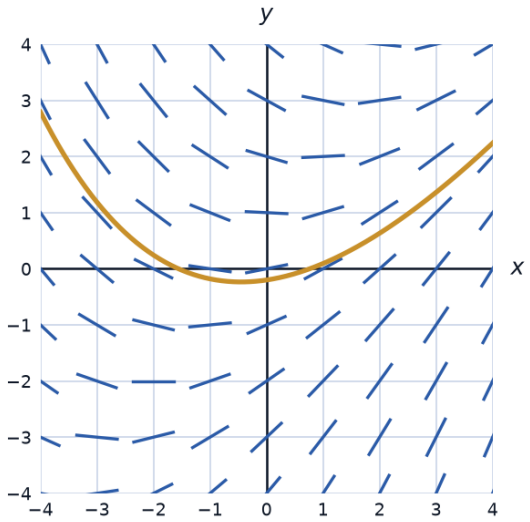
5 [Foundational] • Target: Recognizing and solving separable equations

Solve $\frac{dy}{dx} = xy$.

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6 [Foundational] • Target: Recognizing and solving separable equations

Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.



7 [Foundational] • Target: Recognizing and solving separable equations

Solve $\frac{dy}{dx} = \frac{x+1}{y}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Recognizing and solving separable equations

Solve $\frac{dy}{dx} = 6x^2y^2$.

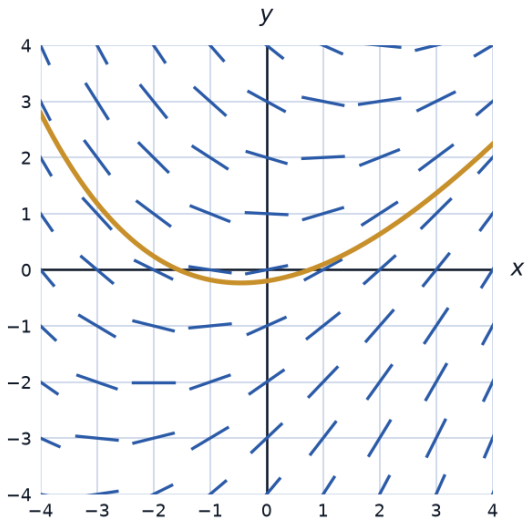
9 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Find the equilibrium (constant) solutions of $\frac{dy}{dx} = y(3 - y)$.

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10 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

For $\frac{dy}{dx} = y(3 - y)$, determine whether $y = 0$ and $y = 3$ are stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below each.



11 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Find the equilibrium solutions of $\frac{dy}{dx} = (y - 2)(y + 1)$ and classify each as stable or unstable.

12 [Intermediate] • Target: General versus particular solutions

Find the general solution of $\frac{dy}{dx} = 2x$, then find the particular solution satisfying $y(1) = 3$.

13 [Intermediate] • Target: General versus particular solutions

Explain, in terms of the arbitrary constant C, why a differential equation typically has infinitely many solutions but an initial value problem typically has exactly one.

14 [Intermediate] • Target: Initial value problems

Solve $\frac{dy}{dx} = 3x^2y$, $y(0) = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

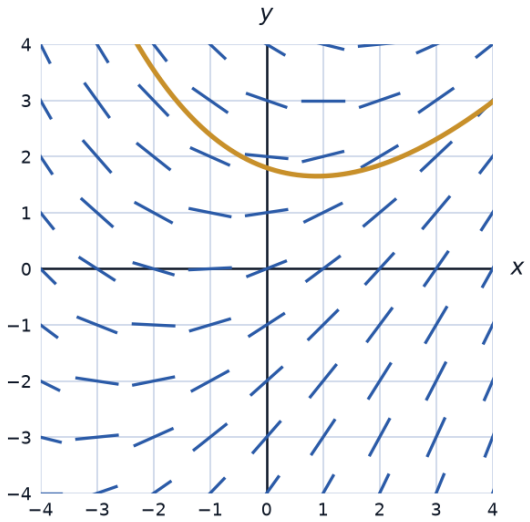
15 [Challenging] • Target: Initial value problems • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve $\frac{dy}{dx} = -2xy$, $y(0) = 5$.

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16 [Challenging] • Target: Initial value problems • Cross-Lesson: 5.01 Variation of Parameters

Solve $\frac{dy}{dx} = 4x^3y$, $y(0) = -3$.



17 [Challenging] • Target: Initial value problems • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y' = -\frac{1}{2}y$, $y(0) = 8$.

18 [Challenging] • Target: Application: radioactive decay and population growth • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k.

19 [Challenging] • Target: Application: radioactive decay and population growth • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.

20 [Challenging] • Target: Application: radioactive decay and population growth • Cross-Lesson: 5.01 Variation of Parameters

A bank account earns continuously-compounded interest, $\frac{dA}{dt} = 0.05A$, with $A(0) = 1000$. Find $A(20)$ and the doubling time.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Separation Form and Integrals

A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is separable if it can be factored into:

a. $\frac{dy}{dx} = g(x)h(y)$

b. $\frac{dy}{dx} = g(x) + h(y)$

c. $\frac{dy}{dx} = g(x/y)$

d. $\frac{dy}{dx} = x^n + y^n$

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22 [Foundational] • Target: Separation Form and Integrals

Separate variables for $\frac{dy}{dx} = 3x^2y^2$.

a. $\frac{1}{y^2}dy = 3x^2dx$

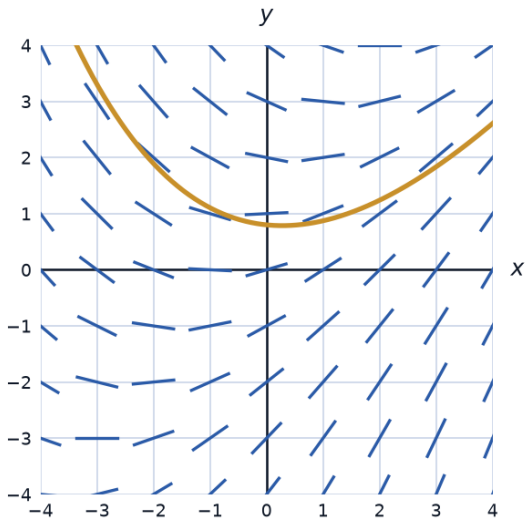
b. $y^2dy = 3x^2dx$

c. $\frac{1}{3x^2}dy = y^2dx$

d. $dy = 3x^2y^2dx$

23 [Foundational] • Target: Separation Form and Integrals

Solve the general equation $\frac{1}{y^2}dy = 3x^2dx$.



a. $y = -\frac{1}{x^3 + C}$

b. $y = \frac{1}{x^3 + C}$

c. $y = -x^3 + C$

d. $y = \sqrt{x^3 + C}$

24 [Foundational] • Target: Initial Value Problems (IVPs)

Solve the IVP: $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = -3$.

a. $y = -\sqrt{x^2 + 9}$

b. $y = \sqrt{x^2 + 9}$

c. $y = -\sqrt{x^2 - 9}$

d. $y = -x - 3$

25 [Foundational] • Target: Initial Value Problems (IVPs)

Solve $\frac{dy}{dx} = 2y$ with $y(0) = 7$.

a. $y = 7e^{2x}$

b. $y = 7e^{-2x}$

c. $y = 2e^{7x}$

d. $y = 7 + 2x$

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26 [Foundational] • Target: Initial Value Problems (IVPs)

What is a singular solution of a differential equation?

a. A solution that cannot be obtained from the general family for any value of the arbitrary constant C

b. A solution that blows up to infinity everywhere

c. A solution where the derivative is undefined

d. A solution with no arbitrary constants

27 [Foundational] • Target: Exponential and Logistic Growth Models

What is the differential equation for logistic population growth with carrying capacity K and growth rate r?

a. $\frac{dP}{dt} = rP\left(1 - \frac{P}{K}\right)$

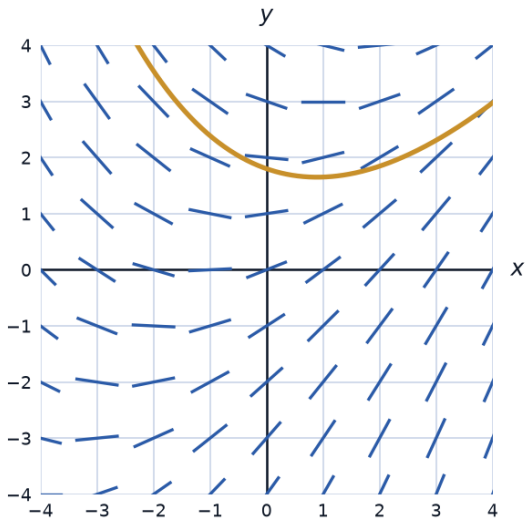
b. $\frac{dP}{dt} = r(K - P)$

c. $\frac{dP}{dt} = rP^2$

d. $\frac{dP}{dt} = \frac{rP}{K}$

28 [Foundational] • Target: Exponential and Logistic Growth Models

At what population size does the logistic growth rate $\frac{dP}{dt}$ reach its maximum?



a. $P = \frac{K}{2}$

b. $P = K$

c. $P = 0$

d. $P = \frac{K}{4}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Exponential and Logistic Growth Models**

What is $\lim_{t\to\infty} P(t)$ in a logistic model if $P(0) > 0$?

a. K

b. ∞

c. 0

d. rK

30 **[Intermediate]** • **Target: Newton's Law of Cooling**

Newton's Law of Cooling states that the rate of change of temperature T is proportional to the difference between T and the ambient temperature T_m:

a. $\frac{dT}{dt} = -k(T - T_m)$ with $k > 0$

b. $\frac{dT}{dt} = -kT$

c. $\frac{dT}{dt} = k(T + T_m)$

d. $\frac{dT}{dt} = -k(T - T_m)^2$

31 **[Intermediate]** • **Target: Newton's Law of Cooling**

A metal bar at 100°C is placed in a room at 20°C. If $T(t) = 20 + 80e^{-kt}$ and cools to 60°C after 10 minutes, find e^{-10k}.

a. $\frac{1}{2}$

b. $\frac{3}{4}$

c. $\frac{1}{4}$

d. $\frac{2}{3}$

32 **[Intermediate]** • **Target: Newton's Law of Cooling**

Using $e^{-10k} = \frac{1}{2}$ from above, what is the temperature after 20 minutes?

a. 40°C

b. 30°C

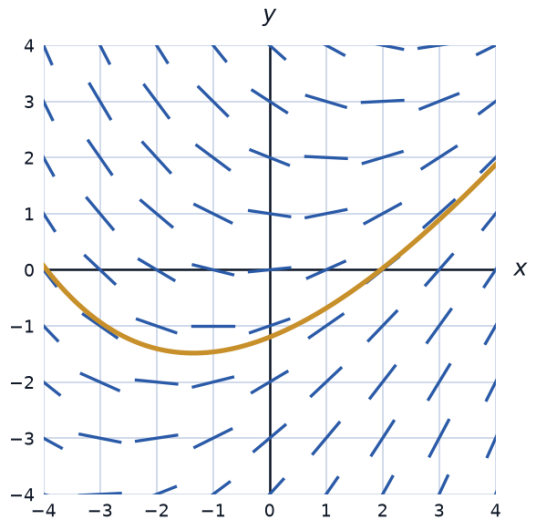
c. 50°C

d. 25°C

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33 [Intermediate] • Target: Newton's Law of Cooling

A hot object cools from 100°C to 80°C in 10 minutes in an environment kept at 20°C. Under Newton's Law of Cooling, at what time t is the instantaneous rate of cooling $|dT/dt|$ exactly half of its initial value at $t = 0$?



a. $10\frac{\ln 2}{\ln(4/3)}$

b. $10\frac{\ln(4/3)}{\ln 2}$

c. 20 minutes

d. $5\frac{\ln 2}{\ln(4/3)}$

34 [Intermediate] • Target: Order and verifying solutions

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.

a. 1

b. 2

c. 6

d. 12

35 [Intermediate] • Target: Recognizing and solving separable equations

An object cools in a room kept at a constant temperature T_a. At $t = 0$, its temperature is 100°C; at $t = 10$ min, it is 60°C; and at $t = 20$ min, it is 40°C. Assuming Newton's Law of Cooling, what is the room temperature T_a?

a. 20°C

b. 15°C

c. 25°C

d. 18°C

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36 [Intermediate] • Target: Equilibrium solutions and qualitative behavior

Hot tea cools from $95\,^\circ\text{C}$ to $80\,^\circ\text{C}$ in 5 minutes in a room kept at $20\,^\circ\text{C}$. Under Newton's Law of Cooling, how many additional minutes will it take to cool from $80\,^\circ\text{C}$ to $50\,^\circ\text{C}$?

a. $10\frac{\ln 2}{\ln(5/4)}$

b. $5\frac{\ln(4/5)}{\ln 2}$

c. $5\frac{\ln 2}{\ln(5/4)}$

d. $5\frac{\ln(5/2)}{\ln(5/4)}$

37 [Intermediate] • Target: General versus particular solutions

A metal ingot at $800\,^\circ\text{C}$ is immersed in a cooling bath kept at $50\,^\circ\text{C}$. After 1 minute, its temperature drops to $500\,^\circ\text{C}$. Under Newton's Law of Cooling, how many additional minutes will it take to cool to $100\,^\circ\text{C}$?

a. $\frac{3\ln 2}{\ln(5/3)}$

b. $\frac{\ln 3}{\ln(5/3)}$

c. $\frac{2\ln 3}{\ln(5/3)}$

d. $\frac{\ln 15}{\ln(5/3)}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Initial value problems • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

An object cools in an environment whose ambient temperature rises linearly according to $T_m(t) = 10 + 2t$ (in $^\circ\text{C}$). Newton's Law of Cooling gives $T' = -k(T - T_m(t))$ with $k = 1\ \text{hr}^{-1}$. If $T(0) = 50\,^\circ\text{C}$, what is $T(1)$?

a. $10 + 42e^{-1}$

b. $10 + 40e^{-1}$

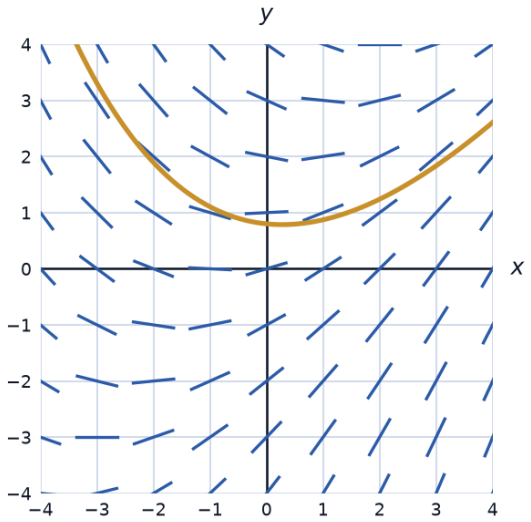
c. $12 + 42e^{-1}$

d. $8 + 42e^{-1}$

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39 [Challenging] • Target: Application: radioactive decay and population growth • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A cup of coffee at $90\,^\circ\text{C}$ is placed in a room held at $20\,^\circ\text{C}$. After 10 minutes, the temperature of the coffee is $60\,^\circ\text{C}$. Assuming Newton's Law of Cooling, at what time t (in minutes) will its temperature reach $30\,^\circ\text{C}$?



a. $10\frac{\ln 7}{\ln(4/7)}$

b. $20\frac{\ln 7}{\ln(7/4)}$

c. $10\frac{\ln 7}{\ln(7/4)}$

d. $10\frac{\ln 4}{\ln(7/4)}$

40 [Challenging] • Target: Application: Newton's Law of Cooling • Cross-Lesson: 5.01 Variation of Parameters

A thermometer initially reading $25\,^\circ\text{C}$ is taken outside into air at constant temperature T_m. After 1 minute it reads $15\,^\circ\text{C}$, and after 2 minutes it reads $9\,^\circ\text{C}$. Assuming Newton's Law of Cooling, what is the outdoor temperature T_m?

a. $-5\,^\circ\text{C}$

b. $5\,^\circ\text{C}$

c. $3\,^\circ\text{C}$

d. $0\,^\circ\text{C}$

41 [Challenging] • Target: Further practice: separable equations • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Under Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_m)$ with surrounding temperature $T_m = 20\,^\circ\text{C}$. An object starts at $T(0) = 80\,^\circ\text{C}$ and cools to $T(1) = 50\,^\circ\text{C}$ at $t = 1$ hour. What is the cooling constant k?

a. $-\ln 2$

b. $\ln 2$

c. $2\ln 2$

d. $\ln(5/8)$

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42 [Challenging] • Target: Mixed review • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the differential equation $\frac{dy}{dx} = \frac{x^2\cos(y)}{1-x^3}$ with the initial condition $y(0) = \frac{\pi}{2}$. If there exists a unique solution in a neighborhood of $x = 0$ and $y = \frac{\pi}{2}$, find the value of $y'(0)$.

a. 0

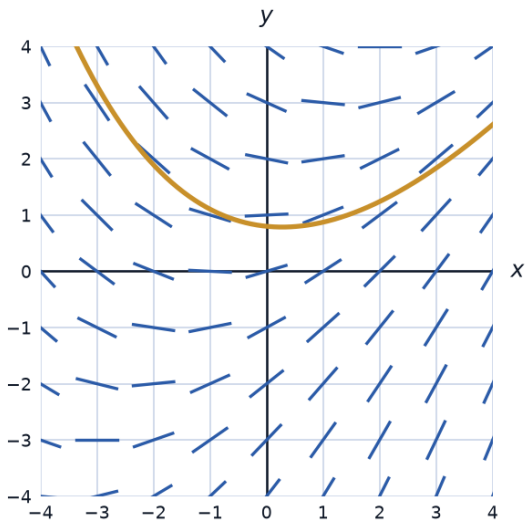
b. 1

c. $\frac{\pi}{2}$

d. Undefined

43 [Challenging] • Target: Separation Form and Integrals • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.



a. $\frac{1}{2}$

b. $\frac{1}{4}$

c. $\frac{1}{2}\sin(2x)$

d. $\frac{1}{4}\cos(2x)$

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44 [Challenging] • Target: Initial Value Problems (IVPs) • Cross-Lesson: 5.01 Variation of Parameters

Consider a second-order linear non-homogeneous differential equation of the form $y''(x) + p(x)y'(x) + q(x)y(x) = g(x)$. If $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions to the corresponding homogeneous equation $y''(x) + p(x)y'(x) + q(x)y(x) = 0$, and the Wronskian is $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$, which of the following expressions represents a particular solution $y_p(x)$ obtained using the method of variation of parameters?

a. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

b. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx + y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

c. $y_p(x) = -y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

d. $y_p(x) = y_1(x)\int \frac{y_2(x)g(x)}{W(y_1,y_2)(x)}\,dx - y_2(x)\int \frac{y_1(x)g(x)}{W(y_1,y_2)(x)}\,dx$

45 [Challenging] • Target: Exponential and Logistic Growth Models • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.

a. $A = 2$

b. $A = 3$

c. $A = 6$

d. $A = 12$

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